# Project 2: Rotor Analysis

## Introduction

BEM (Blade Element Momentum) is a widely used theory in the aerodynamic analysis of rotors. Dr Andrew Ning developed a fast working package called CCBlade.jl for Julia, which I use to run simulations in the APC 10x7E rotor, varying advance ratio and then performing sensitivity tests when varying radius, chord distribution and twist distribution.

## Outcomes

I went over most of the theory with my lab mentor. I learned about the equations that BEM proposes and how it uses control rings and disks to measure momentum of the fluid, which is then used to calculate thrust and torque along the blades of a rotor.

As I followed along the *Quick Start tutorial* and *Guided Examples* I was able to see how the theory gets applied in practice as I wrote the code for this assignment. It also helped me see how the rotor geometry is organized for this kind of analysis: a matrix with a distribution of *radius, chord* and *angle of attack with respect to the direction of freestream velocity.*

Furthermore, I learned the basics of using the PyPlot package, understanding how to plot data properly and how to apply variations to the plot.

## Methods

I created a module, or a separate file with the two functions I needed called Functions.jl. I then created another Main file that would implement such functions and plot the results along with the experimental data.

### The functions:

The first function, called *generate_prop_base* includes the rotor geometry for the APC 10x7E. It does not take any arguments, but it yields rotor, r, chord and theta. Its job is to generate the rotor struct, radial stations, chord distribution, and true twist distribution specifically for the APC 10x7 Thin Electric (10x7E) propeller.

The second function, called *run_simulation*, takes these four arguments along with a range of *advance ratios* and the airfoil distribution (which in this case is an array with the coefficients of lift, drag and moment for the NACA 4412). It then runs the *simple.op* and *solve* function from the CCBlade package, followed by the *thrusttorque* and *nondim* to create arrays with the coefficients and efficiency for the given rotor.

### Varying Radius, Chord distribution and Twist:

I only varied each of these arguments in one way. For example, I performed analysis with only one different twist distribution than the original.

I started with twist because it is the easiest. I simply multiplied each theta by a factor of 1.20 and ran the distribution with this new angle of attack distribution keeping the other variables the same. Chord distribution followed the same idea: I multiplied each chord by 0.80, so the blade is 20% thinner.

For Radius I had to extract the Rtip element in the *rotor* struct using *Rtip_modified = rotor.Rtip * 1.10,* then run the Rotor function again and rescale the radial distance for each r in the rotor geometry matrix (distance from the center to the radial station normalized as r/Rtip). After all this modifications I ran *run_simulation* again to get my plots.

## Results

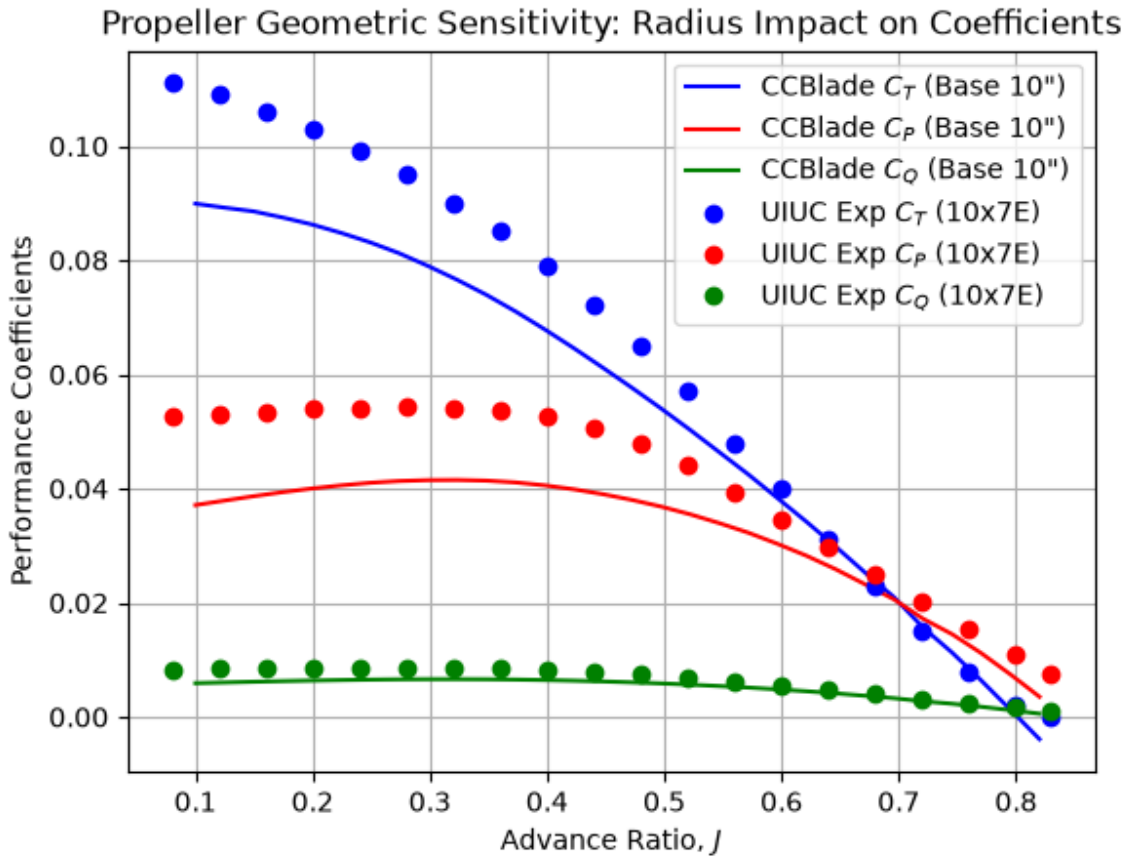
Interestingly, CCBlade analysis did not yield the same results as the experimental data, however, it was presented a close enough distribution, especially for coefficient of torque. And CCBlade suggests a larger efficiency than the experimental data.

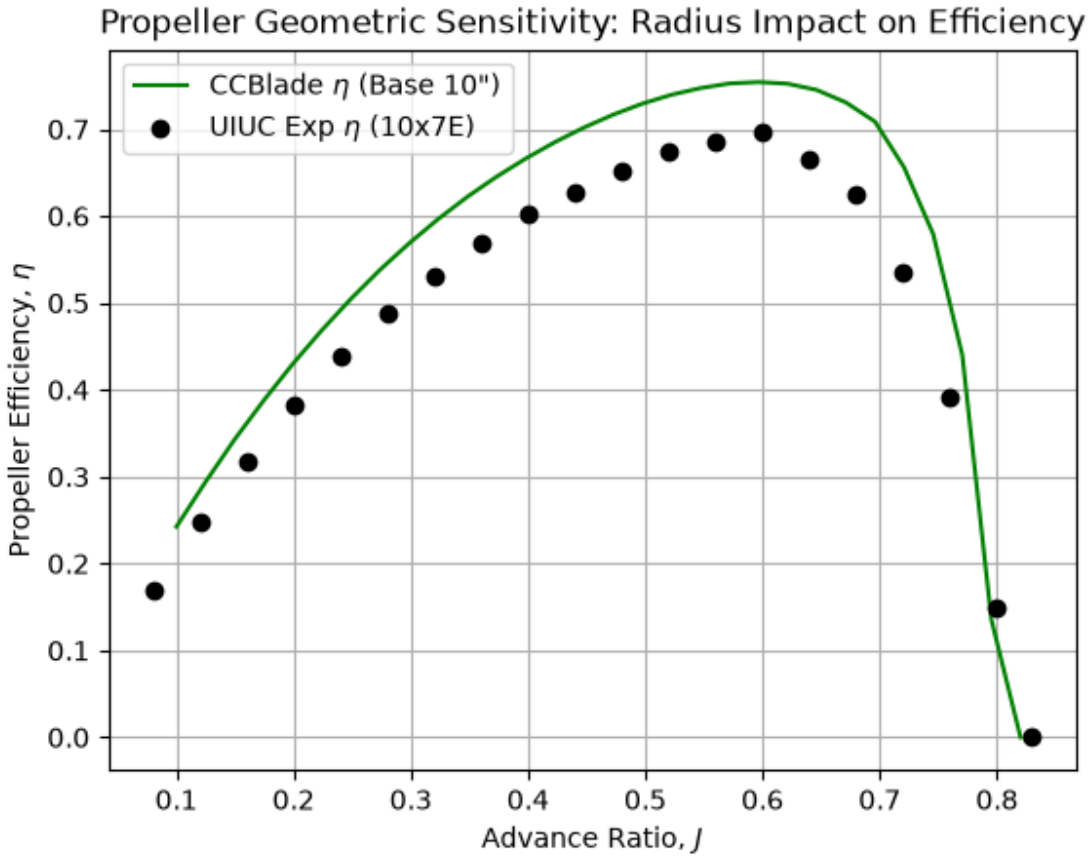
Increasing the twist angle shows a significant increase in all coefficients and a small drop in efficiency.

Making the blades thinner suggests a not significant drop in coefficients, however, it looks like it would cause a small increase in efficiency.

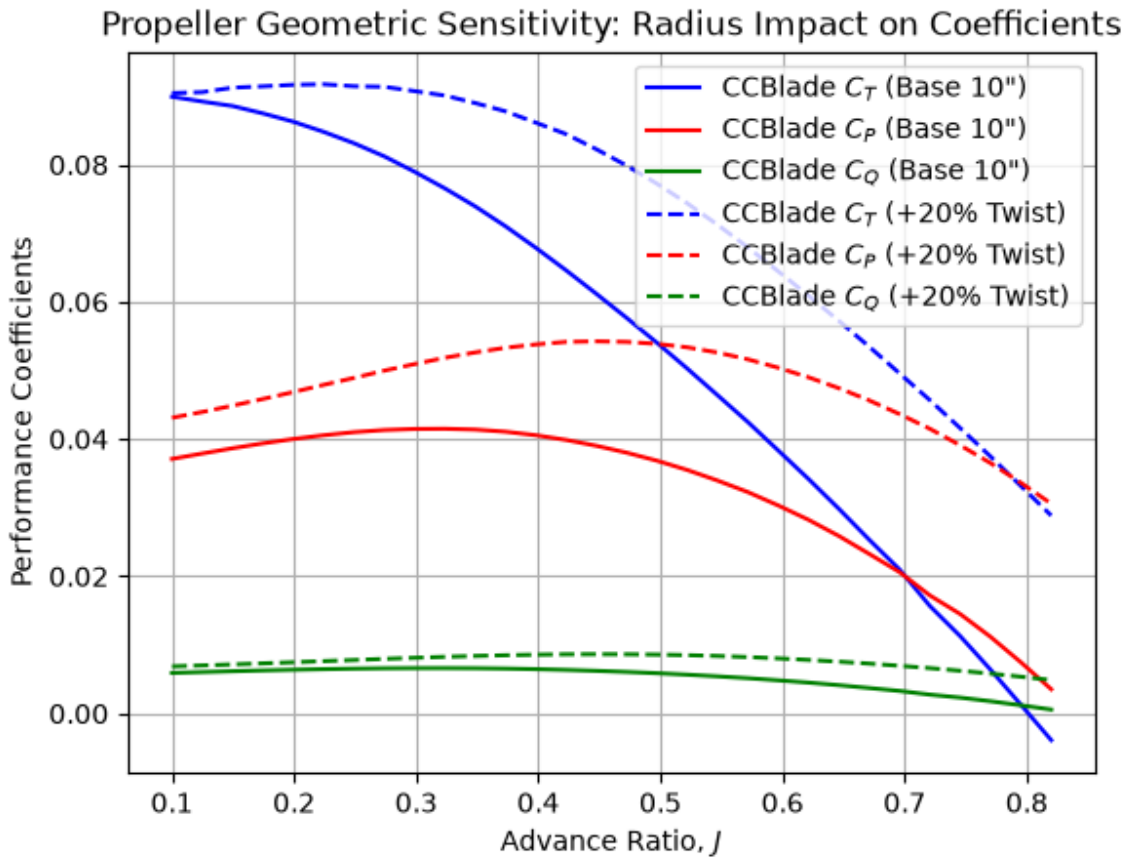
Lastly, a variation in radius shows very little effect on rotor performance

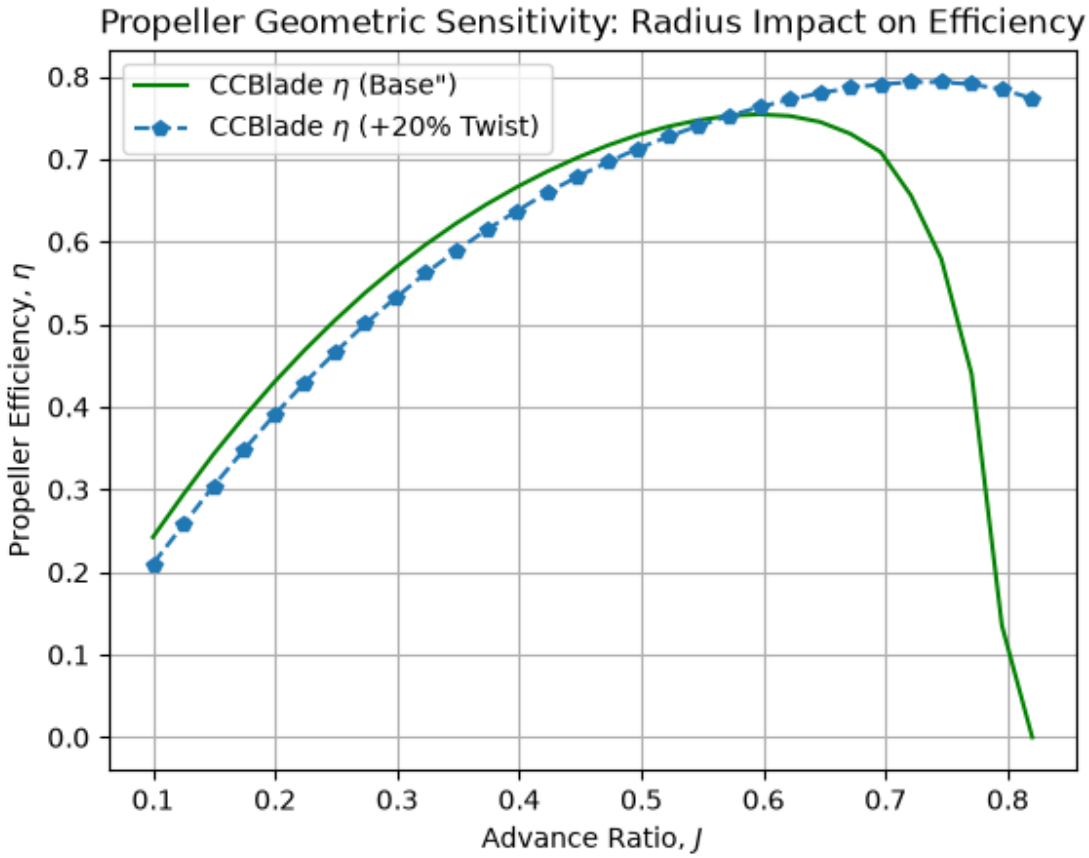
Experimental data vs CCBlade unmodified rotor analysis:



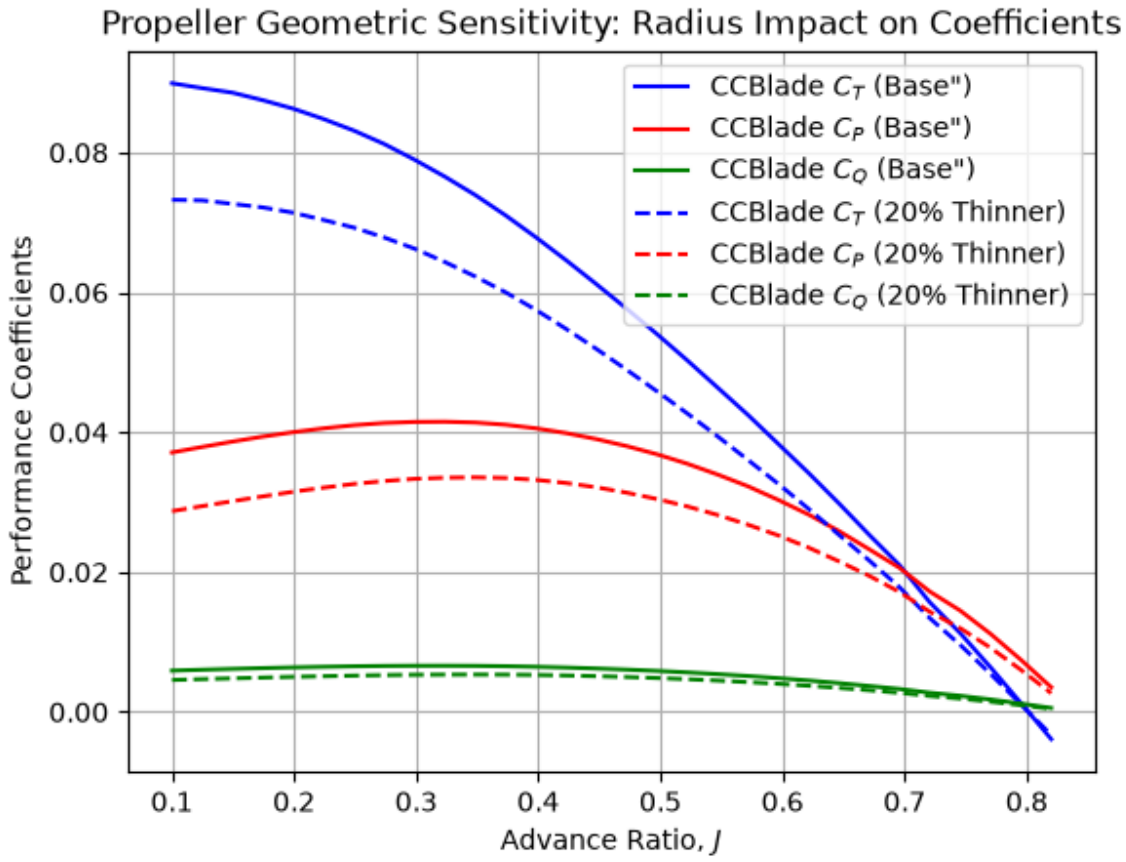

Propeller Geometric Sensitivity: Radius Impact on Efficiency
CCBlade η (Base 10")
UIUC Exp η (10x7E)
Propeller Efficiency, η
Advance Ratio, J
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8

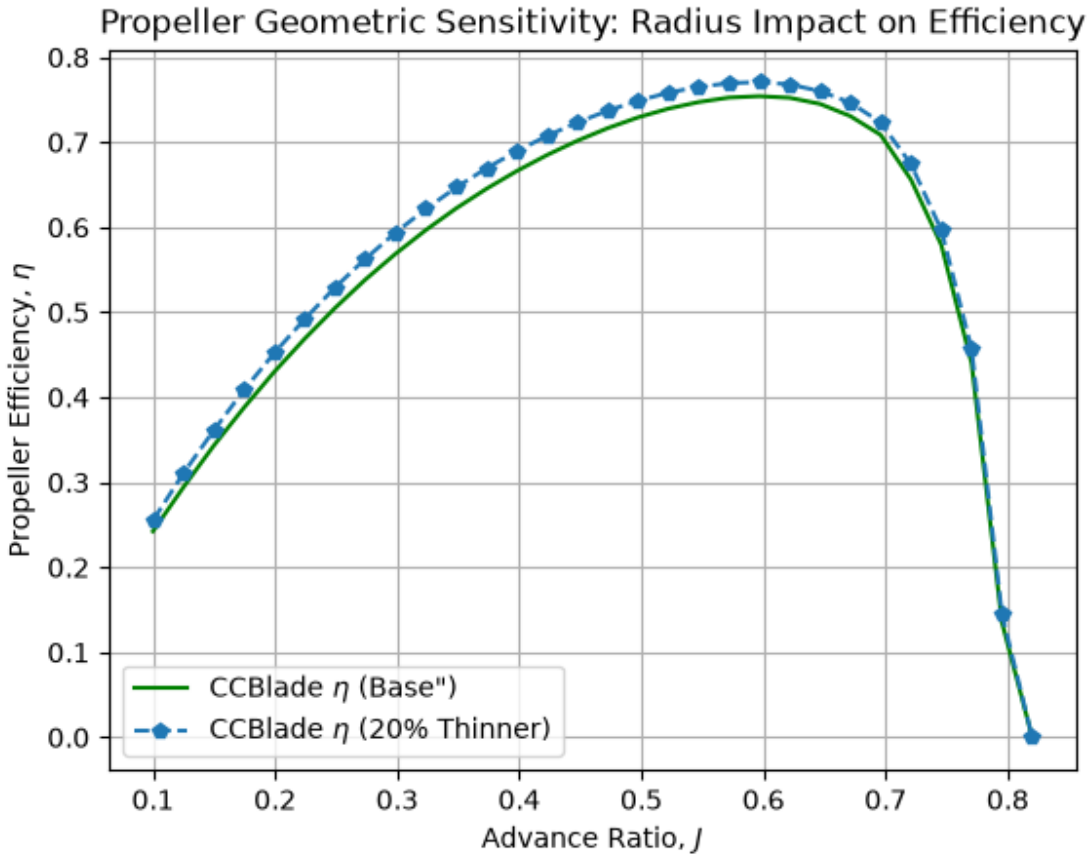
CCBlade analysis of Base rotor vs 20% extra twist rotor



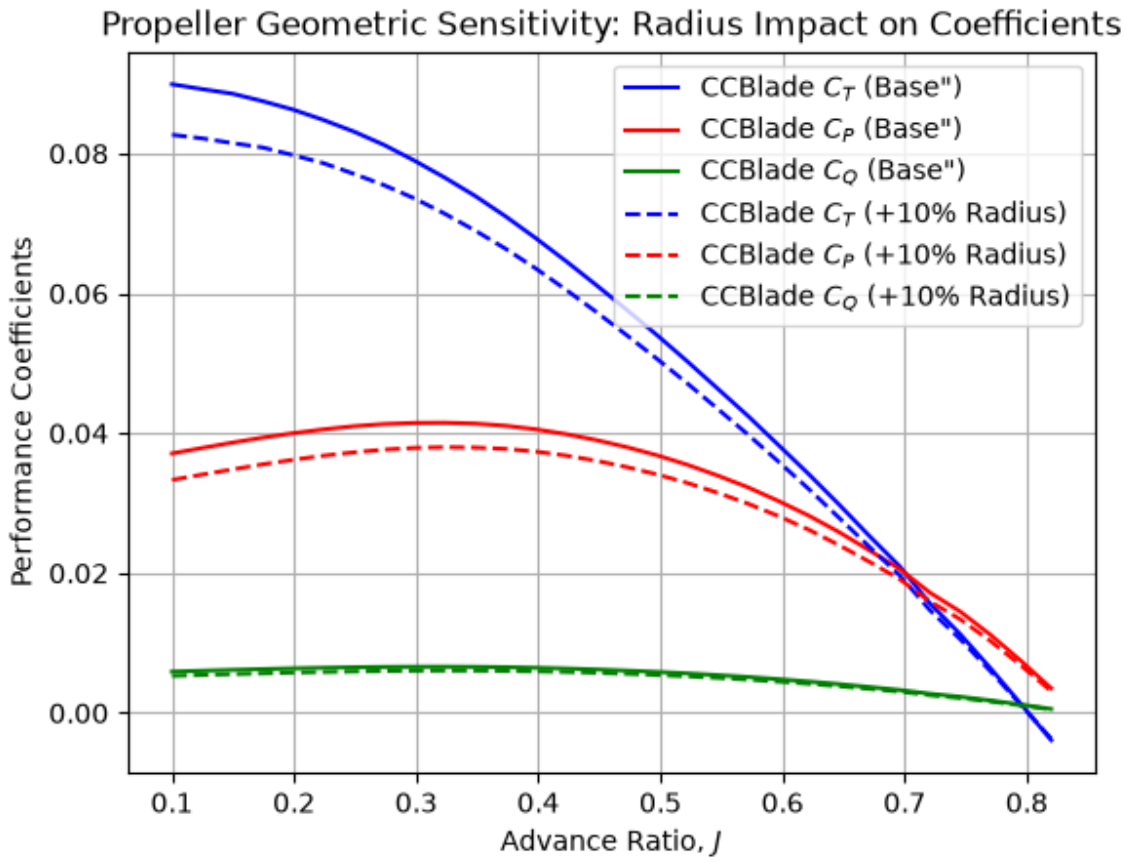

Propeller Geometric Sensitivity: Radius Impact on Efficiency
CCBlade η (Base")
CCBlade η (+20% Twist)
Propeller Efficiency, η
Advance Ratio, J
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8

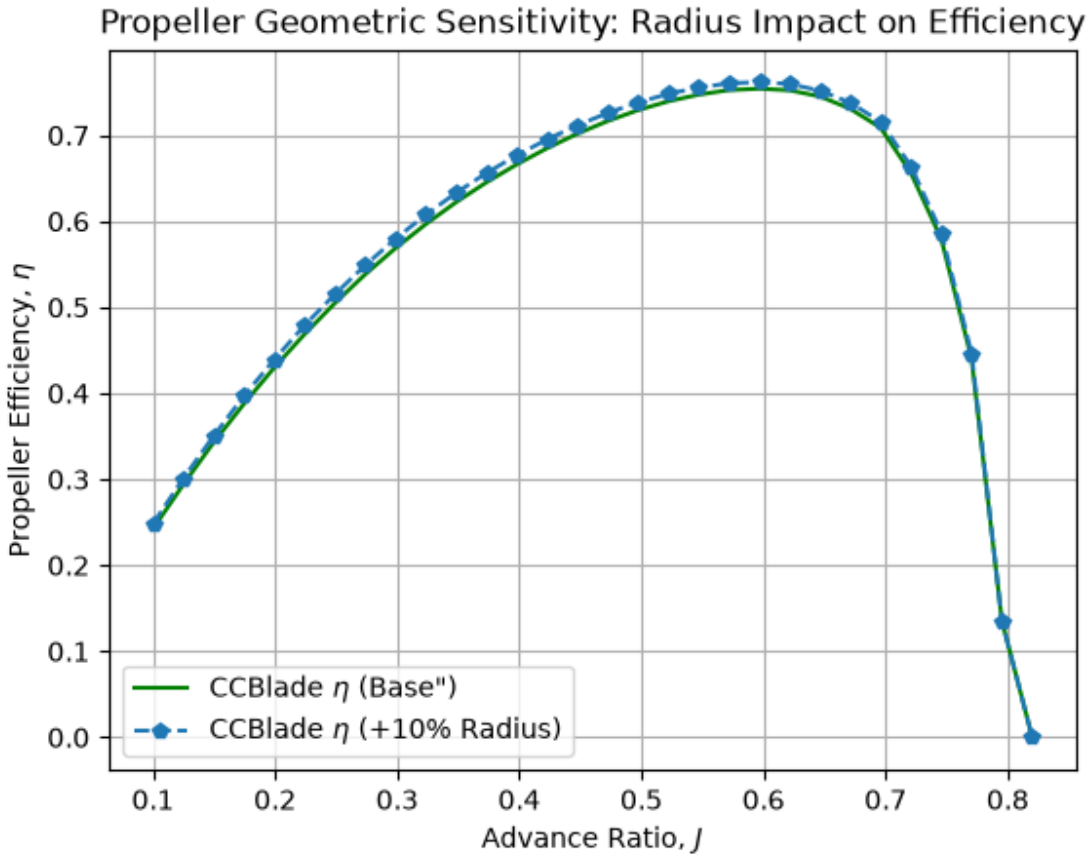
CCBlade analysis of Base rotor vs 20% Thinner blades (varying Chord distribution)




Propeller Geometric Sensitivity: Radius Impact on Efficiency
Propeller Efficiency, η
Advance Ratio, J
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
CCBlade η (Base")
CCBlade η (20% Thinner)

CCBlade Analysis of Base Rotor vs +10% Radius (Radius from the center of hub to the tip of the blade)




Propeller Geometric Sensitivity: Radius Impact on Efficiency
Propeller Efficiency, η
Advance Ratio, J
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
CCBlade η (Base")
CCBlade η (+10% Radius)